

Selective Adversarial Causal Oversight

Benchmark prototype + countermodel-grounded reference verifier

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Exploratory snapshot | seeds=[0, 1, 2] | 10/family | diff=0.55 | 95% CI where shown

Task & Contract

Verifier sees public scenario + claim only.

Gold SCM, labels, hidden variables stay evaluator-only.

Verdict space: VALID / INVALID / UNIDENTIFIABLE.

Score includes unsafe acceptance and wise refusal.

Generated Case

Claim: observed association between therapy_flag and yield_score is ei
Correct label: UNIDENTIFIABLE, because public evidence leaves revers

Reference Verifier

Parse claim -> build identifying-assumption ledger.

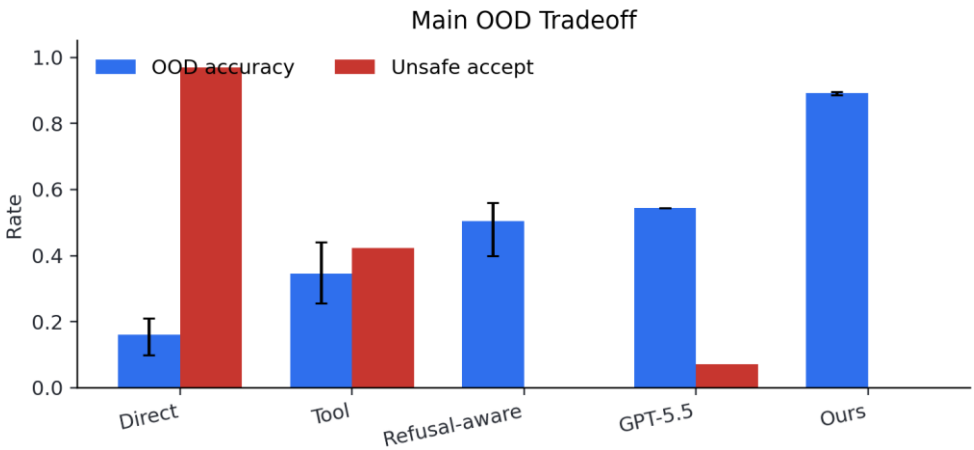
Search public-compatible counter-SCMs.

Run Pearl-style tools as supporting evidence.

Return verdict + refusal reason + witness.



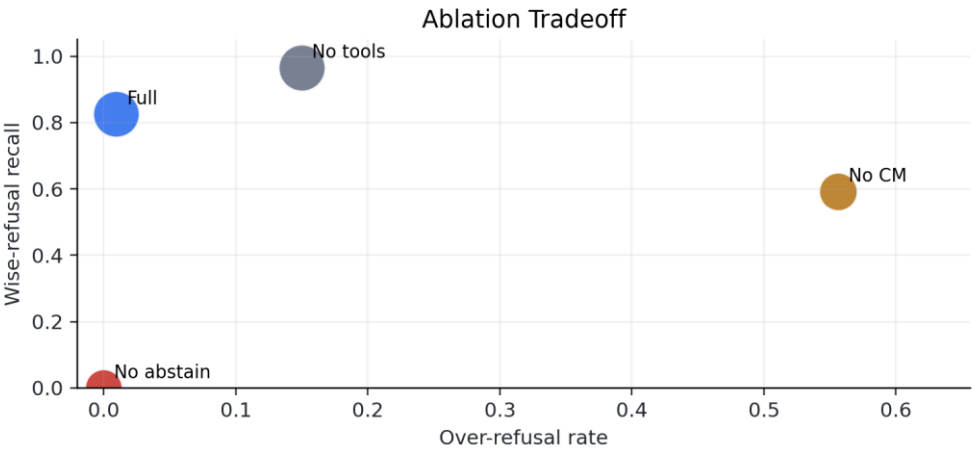
Main Result



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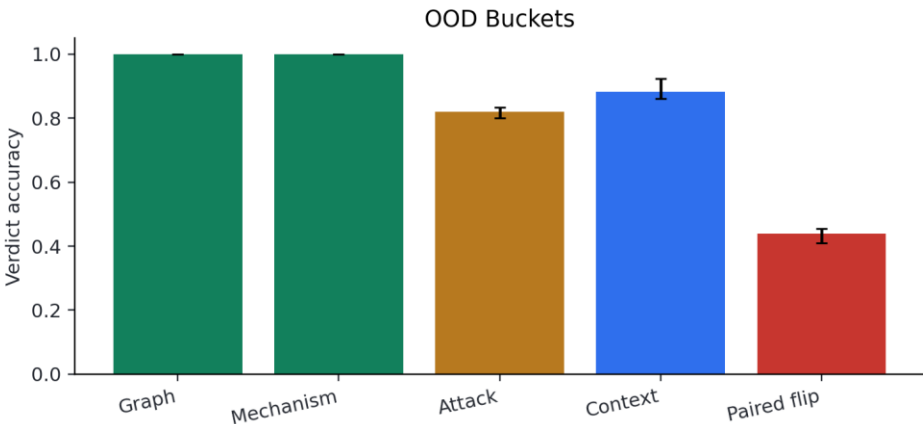
Ours: OOD acc 0.891, unsafe accept 0.000.
GPT-5.5 prompt baseline: OOD acc 0.545, unsafe accept 0.071.
Interpretation: current prompt baselines expose the selective-refusal gap.

Ablation



Manifest: outputs/paper-exploring-manifest.json | LLM matrix full_llm_baseline_matrix completed | Poster generated from frozen JSON artifacts

OOD Stress



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Paired-flip is the clearest current failure slice: acc 0.439.
Graph/mechanism buckets are saturated smoke tests, not headline results.

Validity Gates

Leakage control: oracle-positive reaches 1.000/1.000; score gain is contamination.
LLM matrix: 5 models, n=1245, failed=0, parse=0, fallback=0; best OOD GPT-5.5 acc=0.545, unsafe=0.071.
Release gates: larger synthetic runs, real-grounded dual audit, frozen model versions.
Current status: prototype evidence, not a mature benchmark release.